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Data Analysis Course

The Analysis of Residential Property Pricing: New Cairo vs. Sheikh Zayed City

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Introduction

Egypt's real estate landscape is undergoing a historic transformation, driven by massive urban expansion and the development of world-class satellite cities. For any prospective homeowner or investor, the choice often boils down to a fundamental geographic rivalry: East vs. West. New Cairo and Sheikh Zayed City represent the pinnacles of modern Egyptian living, yet they command significantly different market positions. This report dives into the raw data of thousands of listings to determine if the perceived price gap between these two hubs is a statistical reality or merely a marketing narrative. Understanding these trends is essential for making informed financial decisions in one of the region's most volatile and lucrative sectors.

Research Question

Is there a statistically significant difference in the average residential property prices between New Cairo City and Sheikh Zayed City?

Hypothesis

- Null Hypothesis (H_0): The average property price in New Cairo City is equal to the average property price in Sheikh Zayed City.
- Alternative Hypothesis (H_1): The average property price in New Cairo City is not equal to the average property price in Sheikh Zayed City.

Population of Interest

The population of interest includes all active residential real estate listings and prospective property transactions within the primary urban expansion zones of Greater Cairo, specifically focusing on the luxury and mid-to-high-end residential markets.

Sampling Method

The data was collected via a systematic web-scraping approach from PropertyFinder Egypt, one of the most comprehensive real estate aggregates in the country. This method is the most effective approach for this study as it provides "live" market data that reflects current asking prices, ensuring a high-volume sample that captures the true breadth of the market across various compounds and developers.

Bias Identification

In designing this analysis, we identified **Selection Bias** as a primary concern, as online listings may favor higher-end developments over informal or secondary market sales. To minimize this, we utilized a large-scale dataset and performed rigorous data cleaning. Furthermore, we mitigated **Outlier Bias** by analyzing medians alongside means, ensuring that a few multi-hundred-million EGP villas did not unfairly represent the "typical" home in either city.

The dataset, **"egypt_real_estate_listings.csv"**, contains detailed information on residential listings. The key variables utilized in this analysis are:

- **Location (Independent Variable):** A categorical variable indicating if the property is in New Cairo or Sheikh Zayed.
- **Price (Dependent Variable):** A continuous numerical variable representing the asking price in Egyptian Pounds (EGP).
- **Size:** A continuous numerical variable representing the area of the property in square meters (SQM).
- **Property Type:** Categorical data identifying if the unit is an apartment, villa, or townhouse.

Number of samples used: 6,990 (New Cairo City: 5,124 | Sheikh Zayed City: 1,866)

Analysis

Data Cleaning: Numeric values were extracted by stripping commas from the price column using string replacement and converting to numeric format. To maintain the robustness of the sample size, records with missing values in the cleaned price field were removed using listwise deletion before isolating the two target cities.

City	Mean Price	Median Price	Std. Deviation
New Cairo City	18.9M	11.5M	28.9M
Sheikh Zayed City	13.3M	8.0M	17.2M

Box Plot Showing Residential Property Pricing:

This visualization displays the distribution of prices, highlighting that New Cairo City has a broader interquartile range and a higher median compared to Sheikh Zayed. The chart clearly indicates that while both cities host luxury properties, the "bulk" of the market in New Cairo sits at a significantly higher price point than its western counterpart.



The box plot illustrates that the median price in New Cairo is roughly 3.5 million EGP higher than in Sheikh Zayed, indicating a shift in the central market value."

Furthermore, the significantly larger number of outliers above the 40 million EGP mark in New Cairo confirms its position as the primary hub for ultra-luxury residential listings in Greater Cairo.

Hypothesis Testing Steps

- **Step 1:** Define the Null Hypothesis ($\mathbf{H_0: \mu_1 = \mu_2}$) and Alternative Hypothesis ($\mathbf{H_1: \mu_1 \neq \mu_2}$) at a 0.05 significance level.

- **Step 2:** Select the appropriate test. Based on the project scope and course-approved methods, an **Independent Two-Sample T-test** was chosen.

- **Step 3:** Calculate the test statistics. The analysis yielded a T-statistic of 8.0668 and a P-value of 8.4298×10^{-16}.

Note: Because this number is near zero, we are essentially **99.999%** confident that the price gap between New Cairo and Sheikh Zayed is real and not due to chance.

- Step **4:** Compare the P-value to the significance level ($\boldsymbol{\alpha = 0.05}$). Since the P-value is significantly less than 0.05, we **reject the Null Hypothesis ($\mathbf{H_0}$)**.

Conclusion

The statistical results provide overwhelming evidence to reject the Null Hypothesis. We conclude that there is a statistically significant difference in property pricing between the two cities, with New Cairo City being the more expensive market.

Any potential issues

- **Handling of Missing Values:** One identified issue was the presence of missing price entries. To keep the hypothesis test grounded in verified observations, records with missing cleaned price values were removed through **listwise deletion**, rather than being filled with estimated values. **This approach** maintains consistency with the analysis code and avoids artificially reducing the overall variance of the dataset.

- **Selection Bias:** Because the data is sourced from an online real estate aggregator, there is a natural bias toward modern developments and luxury compounds. Consequently, the dataset may underrepresent older residential units or private sales that do not utilize major digital platforms.

- **Data Integrity:** The accuracy of the findings relies on the information provided by various listing agents. Inconsistencies in how different agents report "total price" versus "down payment" or "installments" can introduce minor noise into the statistical results.